Splay Tree Construction

1 | Algorithms for Splay Tree Construction

To construct a Splay Tree, we perform three major operations:

1. **Splaying:** Moves the accessed node to the root using rotations.
2. **Insertion:** Inserts a new node and then splays it to the root.
3. **Search:** Searches for a node and then splays it to the root.

1.1 | Splay Operation

Algorithm 1 Splay Operation

```
1: function SPLAY(T, X)                          ▷ T is the root, X is the key to splay
2:     if X is root or X is NULL then
3:         return T
4:     end if
5:     if X is left child of parent P then
6:         if P is root then                                          ▷ Zig Rotation
7:             Perform Right Rotation on P
8:         else if X is left child of grandparent G then              ▷ Zig-Zig Rotation
9:             Perform Right Rotation on G
10:            Perform Right Rotation on P
11:        else if X is right child of grandparent G then             ▷ Zig-Zag Rotation
12:            Perform Left Rotation on P
13:            Perform Right Rotation on G
14:        end if
15:    else if X is right child of parent P then
16:        if P is root then                                          ▷ Zig Rotation
17:            Perform Left Rotation on P
18:        else if X is right child of grandparent G then             ▷ Zig-Zig Rotation
19:            Perform Left Rotation on G
20:            Perform Left Rotation on P
21:        else if X is left child of grandparent G then              ▷ Zig-Zag Rotation
22:            Perform Right Rotation on P
23:            Perform Left Rotation on G
24:        end if
25:    end if
26:    return X                                  ▷ X is now at the root
27: end function
```

1.2 | Insertion in Splay Tree

Algorithm 2 Insert Operation

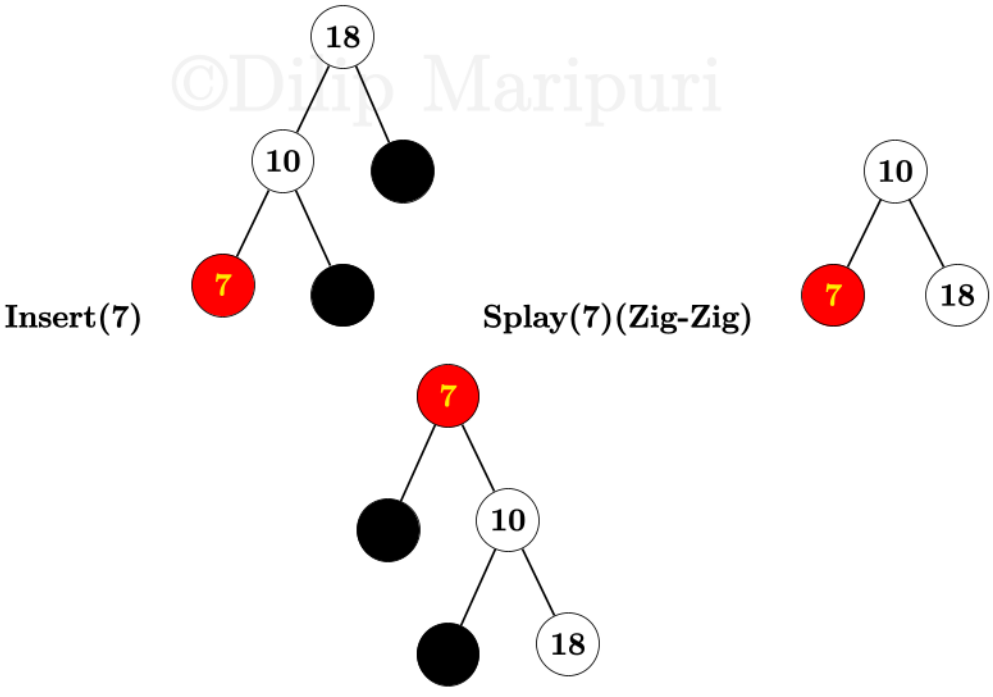
```
1: function INSERT(T, X)                         ▷ T is the root, X is the key to insert
2:     if T is NULL then
3:         Create a new node with key X
4:         return X as root
5:     end if
6:     Perform BST Insertion of X
7:     Splay(T, X)
8:     return X as new root
9: end function
```

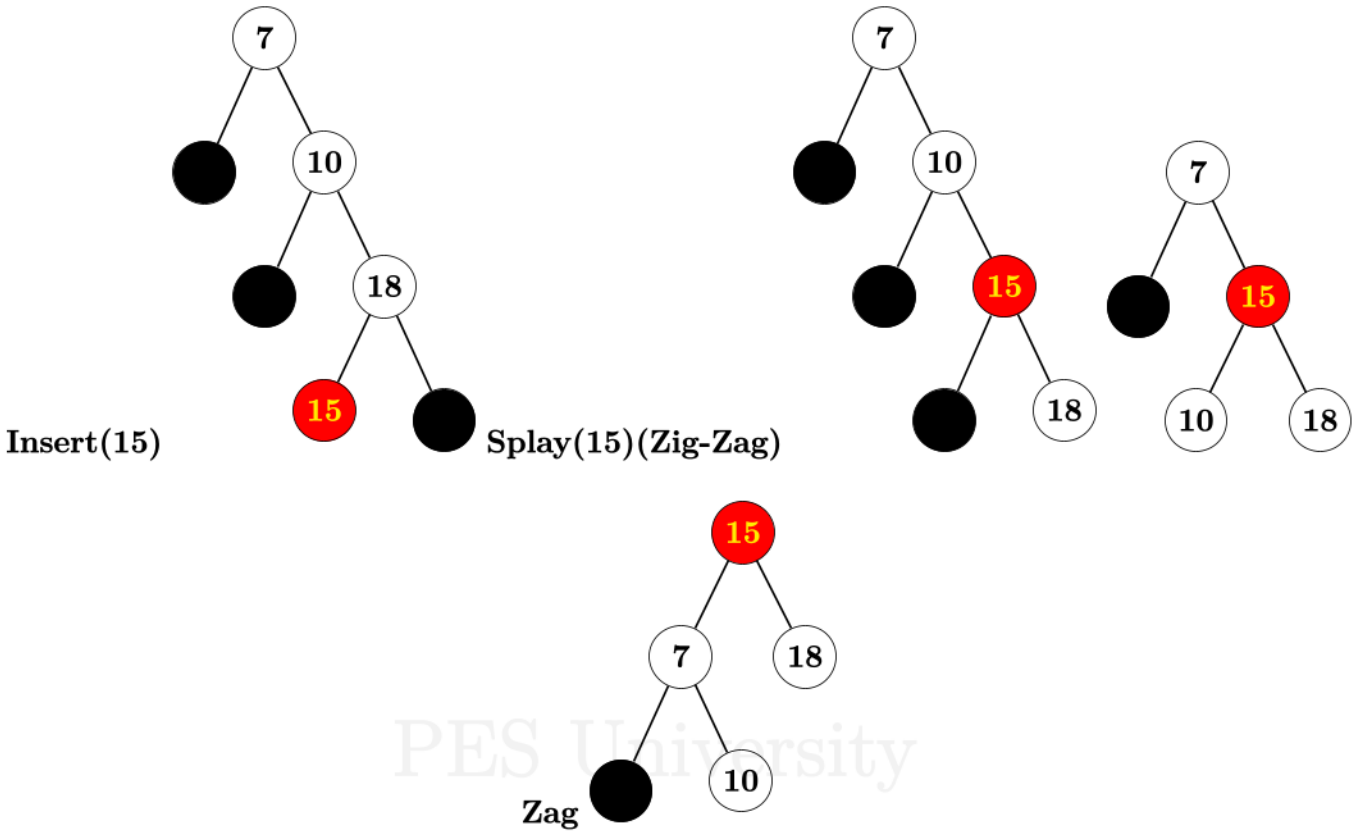
1.3 | Search Operation

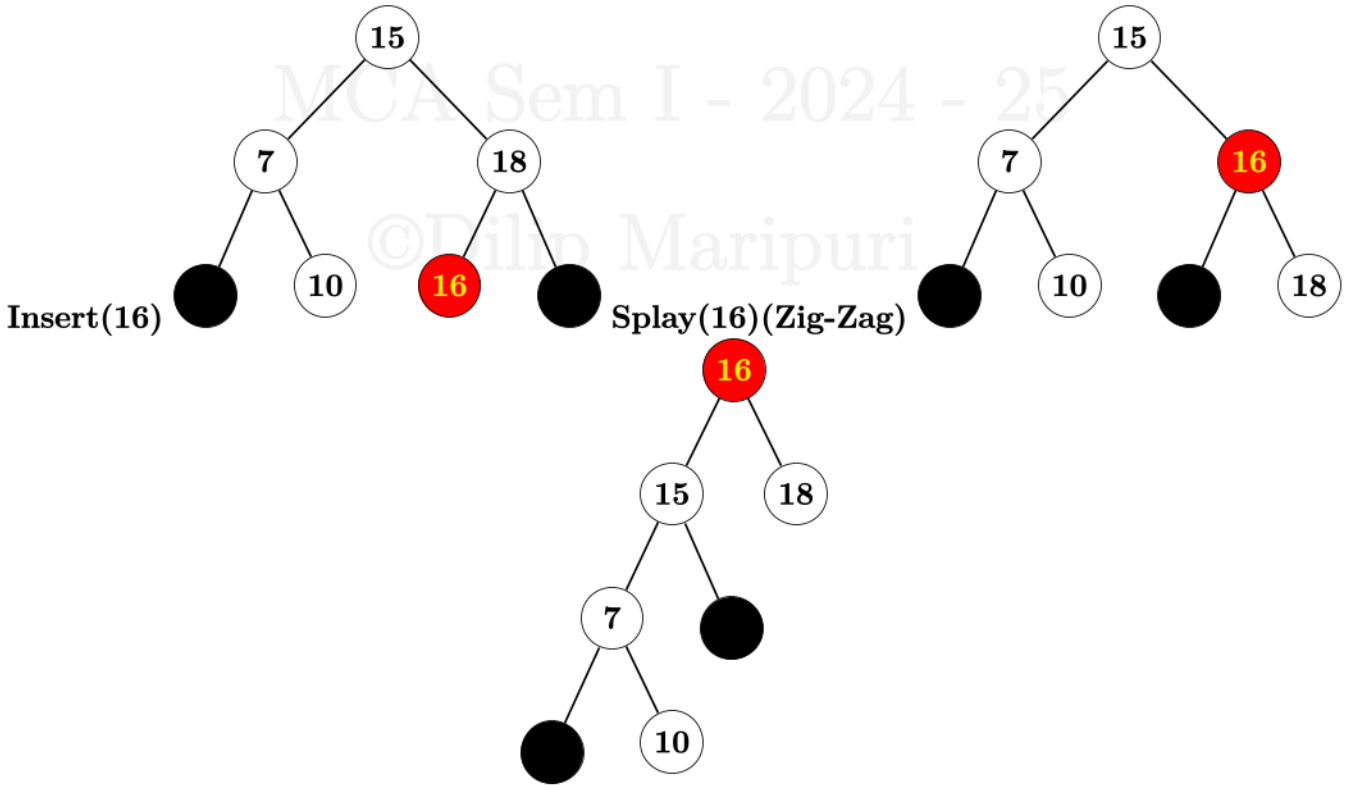
Algorithm 3 Search Operation

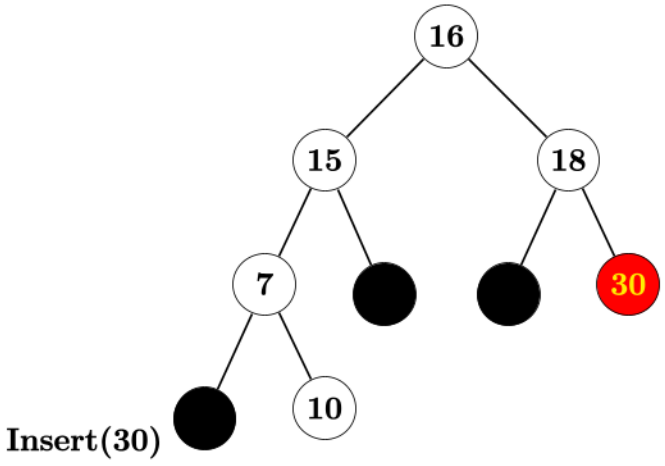
```
1: function SEARCH(T, X)                    ▷ T is the root, X is the key to search
2:     Perform standard BST Search for X
3:     if X is found then
4:         Splay(T, X)
5:     else
6:         Splay(T, last visited node)
7:     end if
8:     return root of updated tree
9: end function
```

2 | Illustration : 10,18,7,15,16,30,25,40,2,70

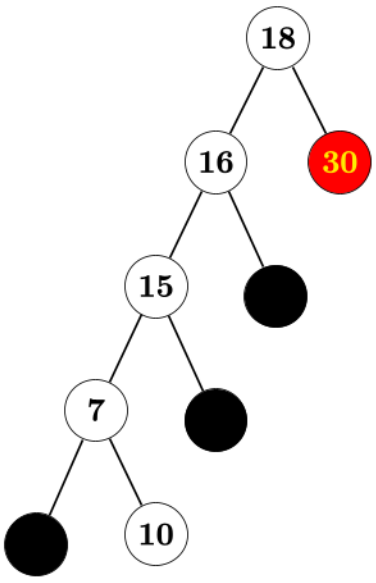




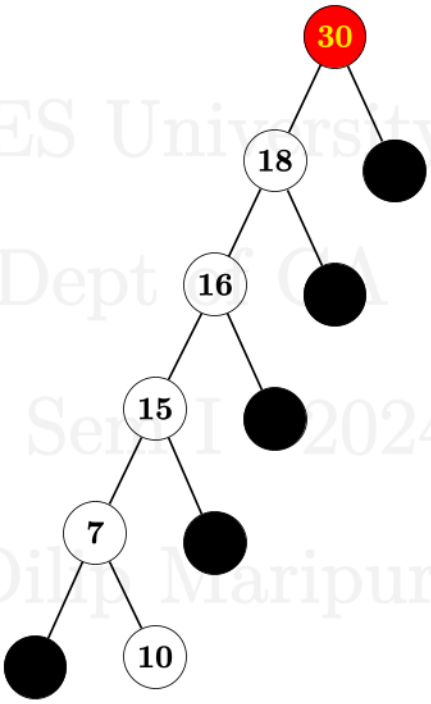


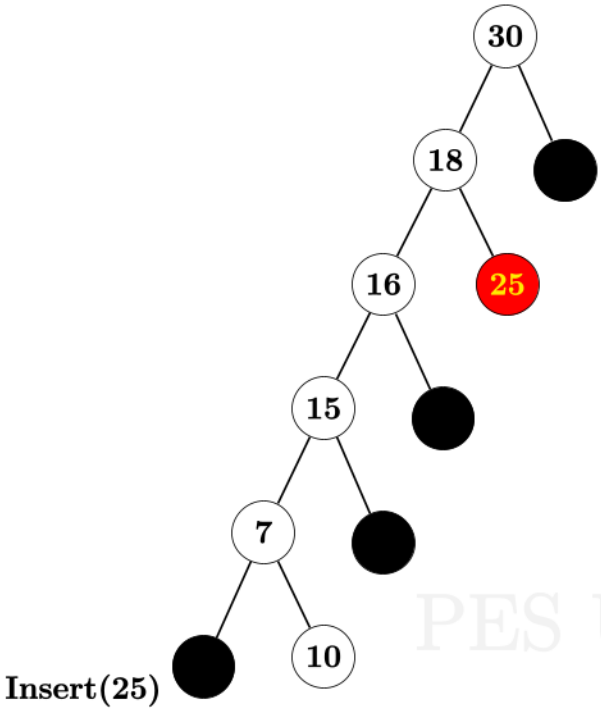


Insert(30)

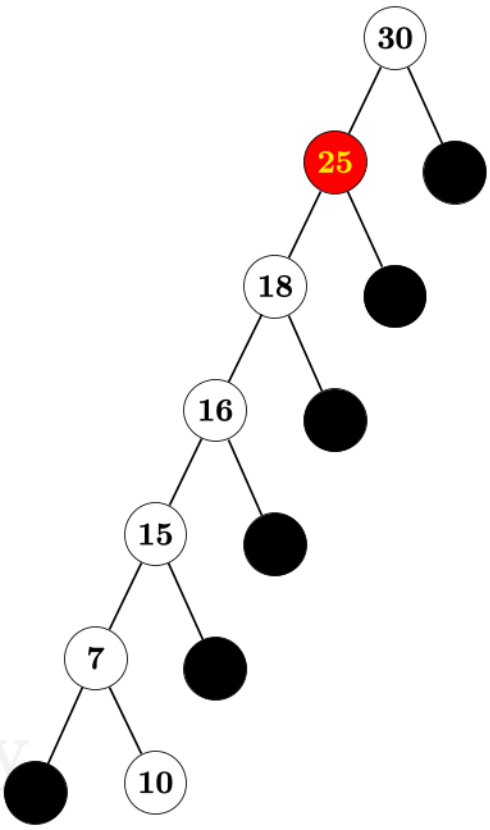


Splay(30)(Zag-Zag)

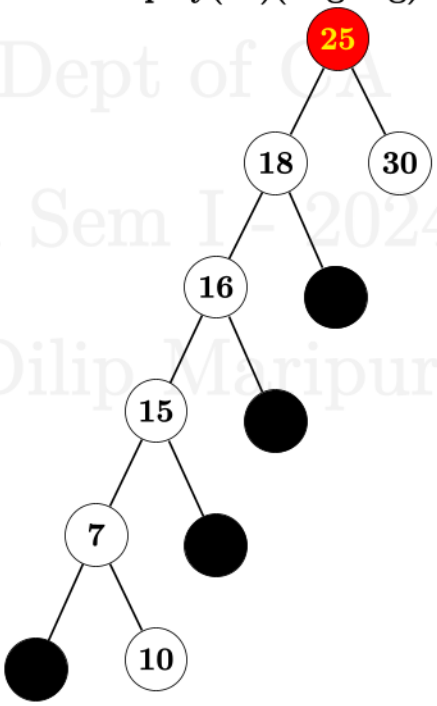


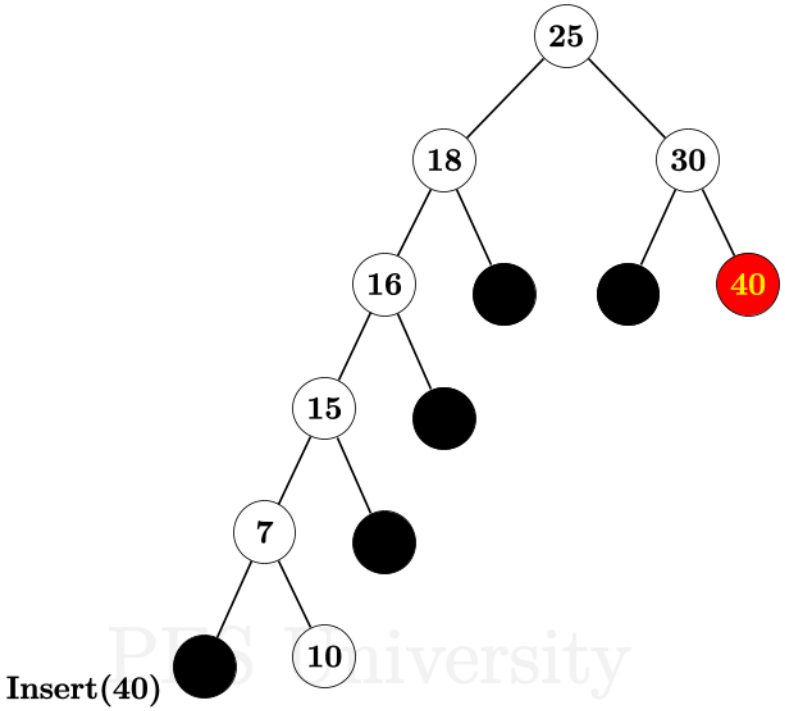


Insert(25)

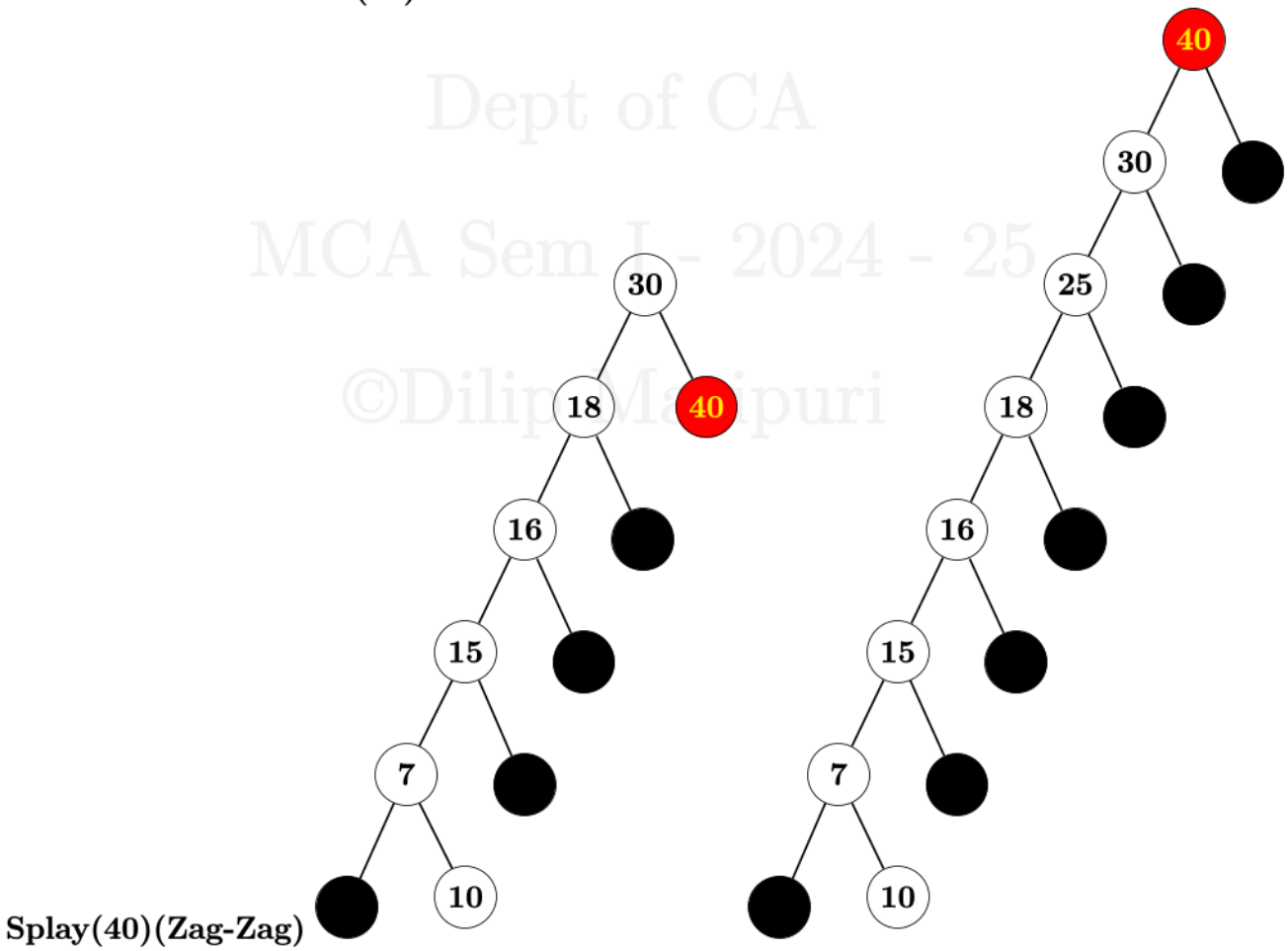


Splay(25)(Zag-Zig)

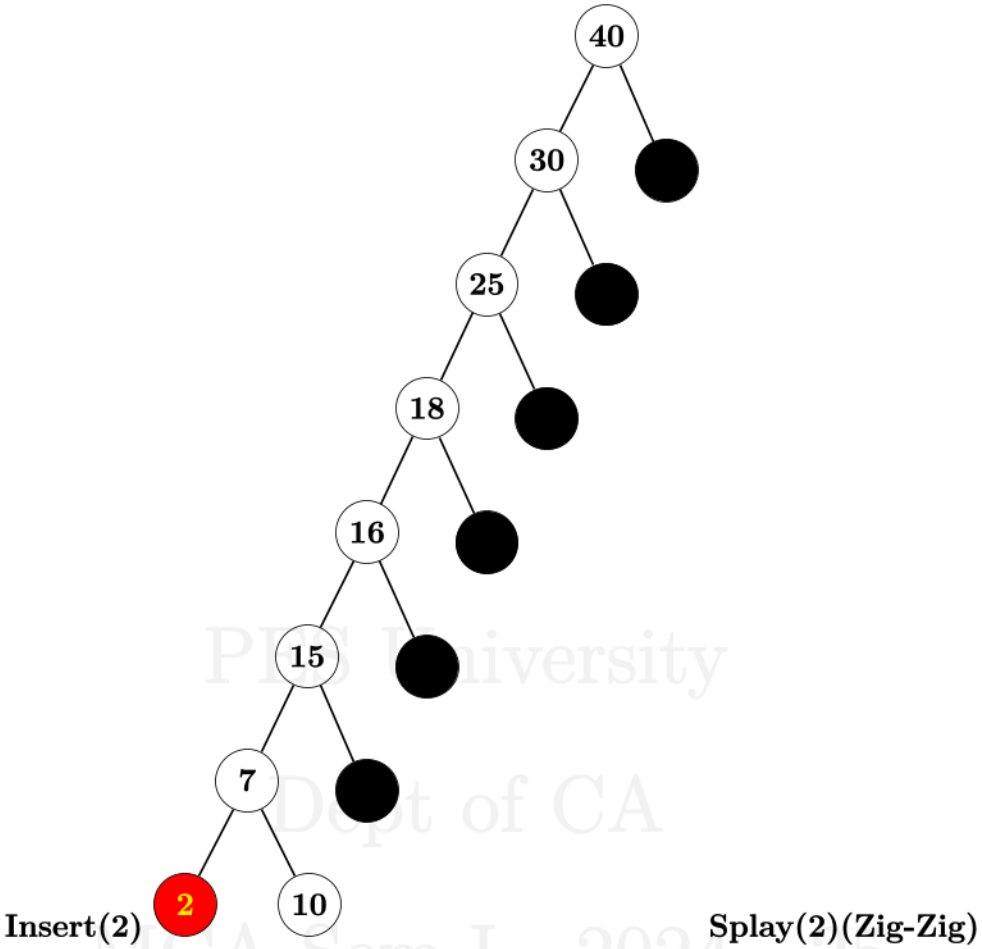


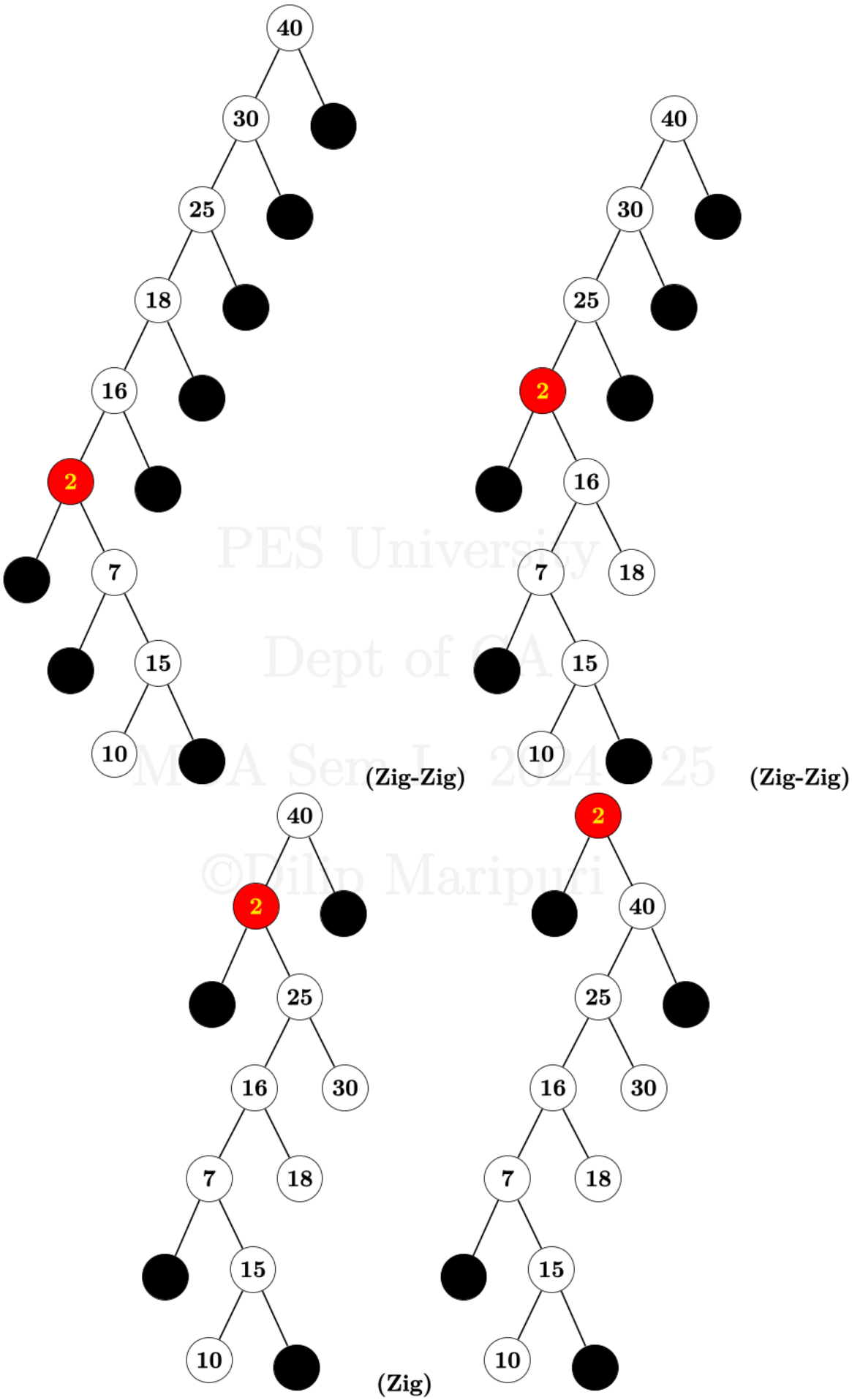


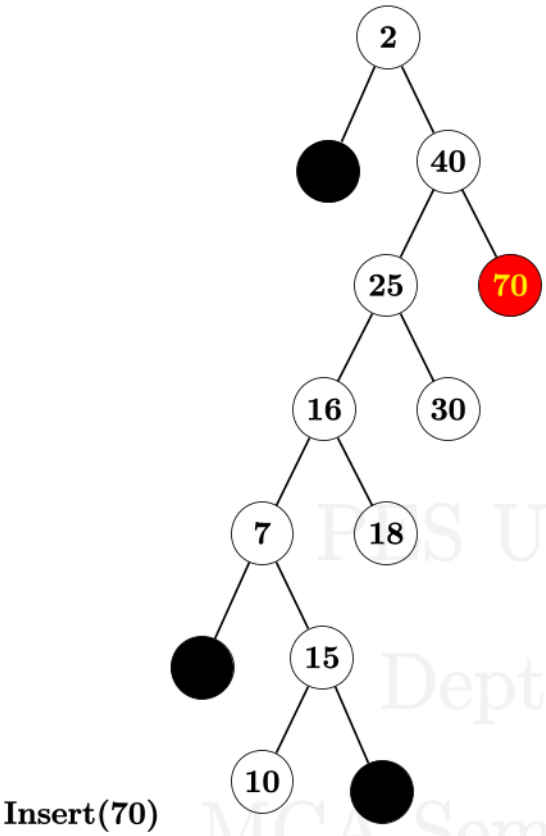
Insert(40)



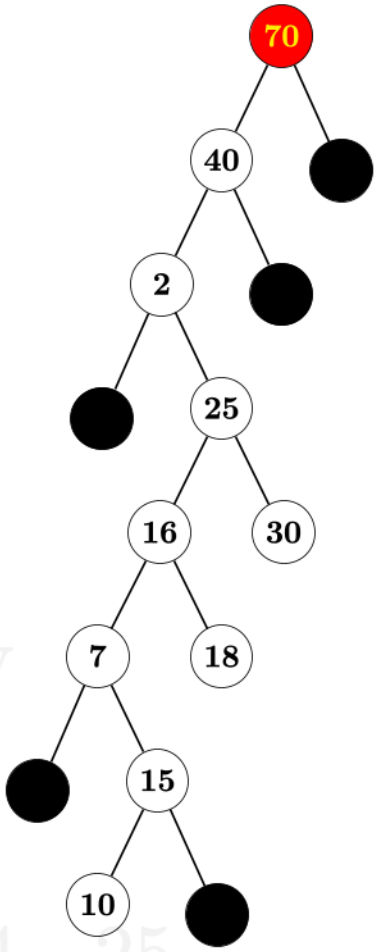
Splay(40)(Zag-Zag)







Insert(70)



splay(70) ZagZag

At the end of Splay Tree Construction, The student should be able to tabulate the following data

Element	Zig	Zag	ZigZig	ZagZag	ZigZag	ZagZig
10						
18		1				
7			1			
15		1			1	
16					1	
30				1		
25						1
40				1		
02	1			3		
70				1		
Total	2	1	1	6	3	1

1. How many transformations are performed to splay 16 after it is inserted? **1-ZigZag**
2. Total No.of transformations that are performed to splay 70 after it is inserted : **14**
3. Is there any transformation that is not applied in the process of construction all the elements into the splay tree in the said order? **NIL**